1. Identity

Document	STAR-PKG-STAR-CROSSCHECK-EN · v2.1 · 2026-11-01
Package	star-crosscheck
Version	0.1.3
Capability	star.crosscheck.fabric
Maturity	RELEASED
Python	⩾3.10
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_crosscheck

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_crosscheck-0.1.3-py3-none-any.whl
```

FILE	SHA-256
star_crosscheck-0.1.3-py3-none-any.whl	510511a948d40a135d10f3ebe1d0a7e1b699c927684c1705ded0c86c233b4c16
star_crosscheck-0.1.3-source.zip	fe771163a691d0d493db8411913ff107e700a8486298c972b8d440877142b3b4

5. What it does, and its limits

Runs engines of declared lineage (each optionally in its own venv) on the same inputs; AGREE only with >= 2 distinct lineages within tolerance, failed engines never count, published references compared; tamper-evident sha256 evidence bundles.

It compares numbers; it does not decide which engine is right when they disagree, and it does not validate any physical model, only agreement within a tolerance the user chooses. Independence is only as good as the declared lineages: an engine described under the wrong lineage (as happened in campaign XC-006, where S.T.A.R.'s own Lambert solver was described as 'universal variables') defeats the independence check, so every lineage is also classified by hand in the registry. Engines that read the same data file are one witness, not two.

- `crosscheck(quantity, inputs, engines, tolerance, ...)` runs every engine (optionally in its own venv via subprocess + JSON) and returns a verdict: AGREE only if at least two DISTINCT lineages agree within the tolerance.
- Engines of the same lineage count as ONE witness (INSUFFICIENT_INDEPENDENCE when fewer than two lineages remain).
- A failed engine never counts as agreement: verdict DEGRADED, with the error recorded.
- Optional published reference: every engine is also compared with it.
- Every result is an evidence bundle with a sha256 over its content (timings excluded) — tamper-evident.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_crosscheck_0_1_3,
  author  = {Cavazzini, Andrea},
  title   = {{star-crosscheck}: Cross-solver verification fabric (S.T.A.R.)},
  version = {0.1.3},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_crosscheck/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts